

1)

```
> db.sinhvien.find({
  phai: "Nu",
  ngoaingu: "Tieng Anh"
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb428b'),
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  tuoi: 20,
  phai: 'Nu',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Phap'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 8
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 7
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb428e'),
  masv: 'sv004',
  hoten: 'Pham Thi Mai',
  tuoi: 21,
  phai: 'Nu',
  malop: '12DHTH',
  ngoaingu: [
```

2)

```
> db.sinhvien.find({
  phai: "Nam",
  tuoi: { $gt: 22 }
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb420c'),
  masv: 'sv002',
  hoten: 'Nguyen Van Tuan',
  tuoi: 23,
  phai: 'Nam',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Nhat'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 9
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 8
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb420d'),
  masv: 'sv003',
  hoten: 'Tran Van Tuan',
  tuoi: 24,
  phai: 'Nam',
  malop: '12DHTH',
  ngoaingu: [
    'Tieng Phap',
    'Tieng Nhat'
```

3)

```

> db.sinhvien.find({
  hoten: { $regex: /^T/ }
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb420b'),
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  tuoi: 20,
  phai: 'Nu',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Phap'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 8
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 7
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb420d'),
  masv: 'sv003',
  hoten: 'Tran Van Tuan',
  tuoi: 24,
  phai: 'Nam',
  malop: '12DHTH',
  ngoaingu: [

```

4)

```
> db.sinhvien.find(
  {
    hoten: { $regex: /Lan$/ }
  },
  {
    _id: 0,
    masv: 1,
    hoten: 1,
    phai: 1
  }
).pretty()
< {
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  phai: 'Nu'
}
```

5)

```
> db.sinhvien.find({
  ngoaingu: {
    $in: ["Tieng Phap", "Tieng Nhat"]
  }
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb420b'),
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  tuoi: 20,
  phai: 'Nu',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Phap'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 8
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 7
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb420c'),
  masv: 'sv002',
  hoten: 'Nguyen Van Tuan',
  tuoi: 23,
  phai: 'Nam',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
```

6)

```
> db.sinhvien.find(
  {
    malop: {
      $in: ["11DHTH", "12DHTH"]
    }
  },
  {
    _id: 0,
    masv: 1,
    hoten: 1
  }
).pretty()
< {
  masv: 'sv001',
  hoten: 'Tran Thi Lan'
}
{
  masv: 'sv002',
  hoten: 'Nguyen Van Tuan'
}
{
  masv: 'sv003',
  hoten: 'Tran Van Tuan'
}
{
  masv: 'sv004',
  hoten: 'Pham Thi Mai'
}
{
  masv: 'sv005',
  hoten: 'Ta Minh Anh'
}
{
  masv: 'sv006',
  hoten: 'Le Van Hung'
}
```

7)

```
> db.sinhvien.find({
  malop: "11DHTH",
  $or: [
    { tuoi: { $lt: 21 } },
    { tuoi: { $gt: 22 } }
  ]
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb420b'),
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  tuoi: 20,
  phai: 'Nu',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Phap'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 8
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 7
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb420c'),
  masv: 'sv002',
  hoten: 'Nguyen Van Tuan',
```

8)

```
> db.sinhvien.find({
  malop: "11DHTH",
  ngoaingu: {
    $in: ["Tieng Phap", "Tieng Nhat"]
  }
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb420b'),
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  tuoi: 20,
  phai: 'Nu',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Phap'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 8
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 7
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb420c'),
  masv: 'sv002',
  hoten: 'Nguyen Van Tuan',
  tuoi: 23,
  phai: 'Nam',
  malop: '11DHTH',
  ngoaingu: [
```

9)

```
> db.sinhvien.find({
  monhoc: {
    $elemMatch: {
      tenmon: "Co so du lieu",
      diem: { $gt: 7.5 }
    }
  }
}).pretty()
< {
  _id: ObjectId('6a8560f955b87d543aeb420b'),
  masv: 'sv001',
  hoten: 'Tran Thi Lan',
  tuoi: 20,
  phai: 'Nu',
  malop: '11DHTH',
  ngoaingu: [
    'Tieng Anh',
    'Tieng Phap'
  ],
  monhoc: [
    {
      mamon: 'm01',
      tenmon: 'Co so du lieu',
      diem: 8
    },
    {
      mamon: 'm02',
      tenmon: 'Lap trinh',
      diem: 7
    }
  ]
}
{
  _id: ObjectId('6a8560f955b87d543aeb420c'),
  masv: 'sv002',
  hoten: 'Nguyen Van Tuan',
```